

Lab: The T-Maze

Learning Goal: Build, run, and analyze the canonical T-Maze Active Inference benchmark.

Part 1: Build the T-Maze

Exercise: Using the code from the module:

```
A, B, C, D = build_tmaze(reward_prob=0.8, cue_reliability=0.9)

# Verify dimensions
for i, a in enumerate(A):
    print(f"A[{i}] shape: {a.shape}, columns sum: {a.sum(axis=0)}")
print(f"B shape: {B.shape}")
print(f"D shape: {D.shape}, sum: {D.sum()}")
```

Verify all matrices are well-formed (columns sum to 1).

Write your response here

Part 2: Run the Agent

Learning Goal: Observe exploration-exploitation behavior.

Exercise: Run 10 trials of the T-Maze and record the agent's behavior:

Trial	Step 1 Action	Step 2 Action	Got Reward?	Visited Cue?
1				
2				
...				

How often does the agent visit the cue before choosing an arm?

Write your response here

Part 3: Parameter Manipulation

Learning Goal: Explore how parameters affect behavior.

Experiments:

1. **Cue reliability = 0.5** (useless cue) — Does the agent still visit the cue?
2. **Cue reliability = 1.0** (perfect cue) — How does behavior change?
3. **C_reward = [0, 0, 0]** (no preferences) — What does the agent do?
4. **C_reward = [10, -10, 0]** (strong preferences) — How does behavior change?

Record results for each experiment.

Write your response here

Part 4: Analyze EFE Components

Learning Goal: Decompose EFE into pragmatic and epistemic terms.

Exercise: For the agent's policy evaluation at the first time step:

```
# Compute EFE for each action and decompose
for action in range(4):
    # YOUR CODE: compute pragmatic and epistemic value
    print(f"Action {action}: Pragmatic={prag:.3f}, Epistemic={epist:.3f},
```

Which action has the highest epistemic value? Why?

Write your response here

Part 5: Reflection

In 150 words, reflect: The T-Maze is a simple environment, yet it demonstrates a profound insight — that curiosity and goal-seeking emerge from one objective (EFE). Is this simplicity a strength or limitation of the benchmark?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	POMDP construction	Multi-modal T-Maze
2	Agent evaluation	Exploration-exploitation behavior
3	Parameter sensitivity	Cue reliability and preference effects
4	EFE decomposition	Pragmatic vs. epistemic value
5	Critical reflection	Benchmark evaluation